

$$A = \begin{pmatrix} S & I & R & D \\ 0{,}95 & 0{,}04 & 0 & 0 \\ 0{,}05 & 0{,}85 & 0 & 0 \\ 0 & 0{,}10 & 1 & 0 \\ 0 & 0{,}01 & 0 & 1 \end{pmatrix}$$

$a_{ij} \in [0,1]$

$\sum_i a_{ij} = 1$

matriz estocástica

$\lambda = 1$ val. pr.

$\forall \lambda \in \sigma(A)$

$|\lambda| \leq 1$

$n = 0 \qquad v(0) = \begin{pmatrix} S(0) \\ I(0) \\ R(0) \\ D(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

$n = 1 \qquad v(1) = A\,v(0)$

$n \qquad v(n) = A^n v(0)$

1) de A diagonalizável

$A = P \cdot D \cdot P^{-1}$

$A^n = P \cdot D^n \cdot P^{-1}$

$v(n) = P \cdot D^n \cdot P^{-1} \cdot v(0) \xrightarrow{n \to \infty} v = \begin{pmatrix} 0 \\ 0 \\ 0{,}(90) \\ 0{,}(09) \end{pmatrix}$

2) $v(n+1) = A\,v(n)$

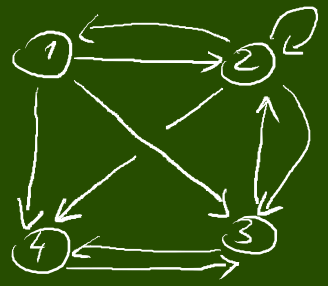
$\exists \lim_{n\to\infty} v(n) = v$

$\longrightarrow v = \begin{pmatrix} 0 \\ 10/11 \\ 1/11 \end{pmatrix}$

$v = Av$

v è vettore proprio per A corrisp. val. pr. $\lambda = 1$

PAGERANK (Google) < Page Brin



$$A = \begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 0 & 1/4 & 0 & 0 \\ 1/3 & 1/4 & 1/2 & 0 \\ 1/3 & 1/4 & 0 & 1 \\ 1/3 & 1/4 & 1/2 & 0 \end{pmatrix} \end{array}$$

$$v(0) = \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{pmatrix}$$

$v(1) = A\,v(0)$

$v(2) = A^2 v(0)$

$v(n) = A^n v(0) \xrightarrow{n \to \infty} ?$

---

1) $A \in \mathcal{M}_n(\mathbb{R})$ stocastică dacă $a_{ij} \in [0,1]$
$A = (a_{ij})$ $\sum_i a_{ij} = 1 \ \forall j$

2) Orice matrice stocastică A are val. pr. $\lambda = 1$
și orice val. pr. $\lambda' \in \sigma(A)$ verifică $|\lambda'| \leq 1$

3) dacă $\exists n \in \mathbb{N}^*$ a.î. $A^n = (b_{ij})$, $b_{ij} > 0$ $\forall i,j$

$\Rightarrow$ a) Val. pr. $\lambda = 1$ este simplă
$a(\lambda) = g(\lambda) = 1 \quad \dim_{\mathbb{R}} V_\lambda = 1$

b) $\forall v(0) \in \mathbb{R}^n$ $\lim_{n \to \infty} \underbrace{A^n v(0)}_{v(n)}$ există
și este independentă de $v(0)$ și este
vect. propriu pt $\lambda = 1$

(thm. PERRON - FROBENIUS)

$\lambda = 1 \Rightarrow v = \begin{pmatrix} 0,1875 \\ 0,75 \\ 1 \\ 0,75 \end{pmatrix} = \begin{pmatrix} 3/16 \\ 3/4 \\ 1 \\ 3/4 \end{pmatrix}$

$v = \alpha \cdot \begin{pmatrix} 3/16 \\ 3/4 \\ 1 \\ 3/4 \end{pmatrix} \qquad \alpha \left(\frac{3}{16} + \frac{3}{4} + 1 + \frac{3}{4} \right) = 1$

$\alpha = \frac{16}{43}$

$\lim_{n \to \infty} A^n v(0) = \begin{pmatrix} 3/43 \\ 12/43 \\ 16/43 \\ 12/43 \end{pmatrix}$

$0,8 - 0,9$

$d = 0,85$ damping factor

$v(n+1)_i = d \sum_{j=1}^{4} a_{ij} v(n)_j + (1-d) \frac{1}{4} \sum_{j=1}^{4} v(n)_j \qquad i = \overline{1,4}$

$= \left(d \cdot A + (1-d) \cdot \frac{1}{4} \begin{pmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{pmatrix} \right) v(n)$

$\tilde{A} = (\tilde{a}_{ij}) \qquad \tilde{a}_{ij} > 0$

$\lambda = 1$

$\lim_{n \to \infty} v(n) \ \exists$ indep de $v(0)$

$k = \mathbb{R}$ $\quad (k = \mathbb{C})$

$V \times V \longrightarrow \mathbb{R}$ $\mathbb{C}$

$(u, v) \longrightarrow \dfrac{\langle u, v \rangle}{u \cdot v}$

Quantum

Hermite

$\langle u, v \rangle = \overline{\langle v, u \rangle}$

$\mathbb{R}^n \qquad u, v \qquad \langle u, v \rangle = \sum_{i=1}^{n} x_i y_i$

1) simetria $\langle u, v \rangle = \langle v, u \rangle$

2)+ 3) liniaritatea $\langle u_1 + u_2, v \rangle = \langle u_1, v \rangle + \langle u_2, v \rangle$

$\langle u_1 + u_2, v \rangle = (u_1 + u_2)^t \cdot v = (u_1^t + u_2^t) \cdot v$

$= u_1^t v + u_2^t v = \langle u_1, v \rangle + \langle u_2, v \rangle$

$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$

$\langle \lambda u, v \rangle = (\lambda u)^t v = \lambda u^t v = \lambda \langle u, v \rangle$

4) + 5) pozitywność : $\langle u, u \rangle \geq 0$

$$\langle u, u \rangle = \sum_{i=1}^{n} x_i^2 \geq 0$$

$$u = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$\langle u, u \rangle = 0 \iff x_i = 0 \ \forall i \iff u = 0$

$$\mathbb{R}_n[x] \simeq \mathbb{R}^{n+1}$$

$$P(x) \longrightarrow \begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix}$$

$P(x) = a_0 + \ldots + a_n x^n$

$$\langle f, g \rangle = \int_a^b f(t) g(t) \, dt$$

$$\langle f, f \rangle = \int_a^b f^2(t) \, dt \geq 0$$

$$\langle f, f \rangle = 0$$

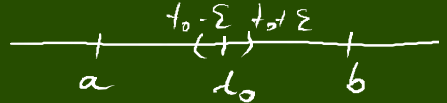
$$\int_a^b f^2(t)\,dt = 0 \xrightarrow{?} f(t) = 0 \quad \forall t \in [a,b]$$

f contínua

pp. $\exists\, t_0 \in (a,b)$ a. i. $f(t_0) \neq 0$

pp $f(t_0) > 0 \Rightarrow \exists \varepsilon > 0$ $f(t) > 0$

$t \in (t_0 - \varepsilon, t_0 + \varepsilon)$

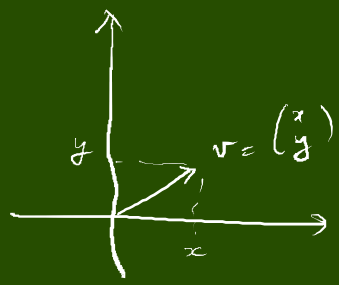


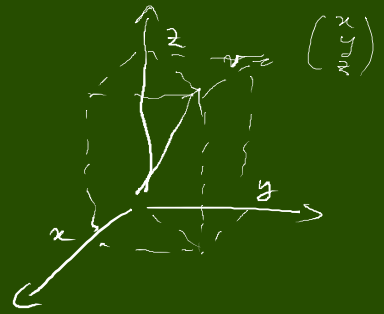
$$0 = \int_a^b f^2 = \int_a^{t_0-\varepsilon} f^2 + \int_{t_0-\varepsilon}^{t_0+\varepsilon} f^2 + \int_{t_0+\varepsilon}^{b} f^2$$

$[a, t_0-\varepsilon]$: ≥ 0 ; $[t_0-\varepsilon, t_0+\varepsilon]$: > 0 ; $[t_0+\varepsilon, b]$: ≥ 0

$\Rightarrow$ pp. falsa $\Rightarrow$ $f(t) = 0 \quad \forall t \in [a,b]$

$\|v\| = \sqrt{\langle v, v \rangle}$ $\quad |v|$





$\|v\| = \sqrt{x^2+y^2}$ $\qquad \|v\| = \sqrt{x^2+y^2+z^2}$

$\mathbb{R}^n \qquad u = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad v = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$

$$(x_1y_1 + \dots + x_ny_n)^2 \leq (x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2)$$

$(V, \langle , \rangle)$ sp. euclidiano

$$\langle u, u \rangle \geq 0$$

$$\langle u + \lambda v, u + \lambda v \rangle \geq 0 \qquad \forall \lambda \in \mathbb{R}$$

$$\langle u + \lambda v, u + \lambda v \rangle = \langle u, u + \lambda v \rangle + \lambda \langle v, u + \lambda v \rangle$$

$$= \langle u + \lambda v, u \rangle + \lambda \langle u + \lambda v, v \rangle$$

$$= \langle u, u \rangle + \lambda \langle v, u \rangle + \lambda \langle u, v \rangle + \lambda^2 \langle v, v \rangle$$

$$= \underbrace{\lambda^2 . \|v\|^2 + 2\lambda \langle u, v \rangle + \|u\|^2}_{\geq 0 \qquad \forall \lambda \in \mathbb{R}}$$

$$\Rightarrow \quad \Delta \leq 0$$

$$\Delta = 4 \langle u, v \rangle^2 - 4 \|u\|^2 . \|v\|^2 \leq 0$$

$$\Rightarrow \quad \langle u, v \rangle^2 \leq \|u\|^2 . \|v\|^2$$

$$|\langle u, v \rangle| \leq \|u\| . \|v\|$$

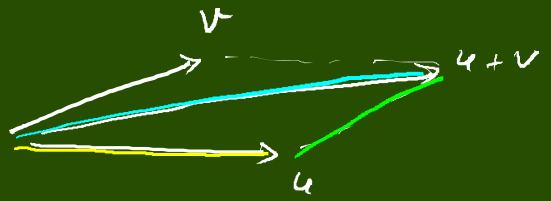
$$\|u+v\| \leq \|u\| + \|v\|$$

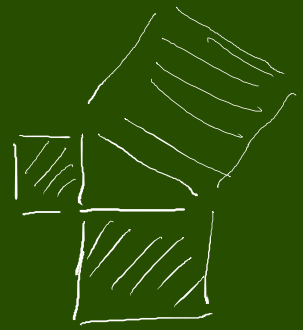
$$\|u+v\|^2 = \langle u+v, u+v \rangle =$$

$$= \|u\|^2 + 2\langle u, v \rangle + \|v\|^2$$

$$\leq \|u\|^2 + 2\|u\| \cdot \|v\| + \|v\|^2$$

$$= (\|u\| + \|v\|)^2$$



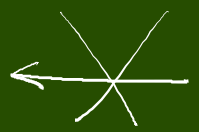


$\langle u, v \rangle = 0$ (u, v ortogonali)

$$\Leftrightarrow \quad \|u+v\|^2 = \|u\|^2 + \|v\|^2$$



$(V, \langle , \rangle) \longrightarrow$ sp. vectorial normat $(V, \|\cdot\|)$



exemplu. $V = \{ f : [a,b] \to \mathbb{R} \mid f \text{ cont} \}$

$\|f\|_\infty = \sup_{x \in [a,b]} |f(x)| = \max_{x \in [a,b]} |f(x)|$

$|x+y| \leq |x| + |y|$

$\nexists \ \langle f, g \rangle$ a.i. $\|f\| = \sqrt{\langle f, f \rangle}$

$\langle u+v, u+v \rangle$

Teorema 1) $(V, \langle , \rangle) \Rightarrow \|u+v\|^2 + \|u-v\|^2 =$

$= 2(\|u\|^2 + \|v\|^2)$ (rel. paralelogr.)

2) $f : [0,1] \to \mathbb{R}$ $f(x) = x$

$g : [0,1] \to \mathbb{R}$ $g(x) = 1$

$$\|f+g\|^2+\|f-g\|^2 \neq 2(\|f\|^2+\|g\|^2)$$

